DPUTRFX

DPUTRFX performs the simulation of road users and their behaviour models. Additionaly, DPUTRFX acts as a network server fort the instances of DPUTRFClient on other computers of the simulator network (e.g. image generators).

DPUTRFX can only run together with DPUSCNX.

1. Inputs:

ControllerTrigger	If this parameter is set to *n*, then every *n*th model will be triggerd in each simulation step. You can increase this parameter to reduce calculation time of the DPUTRFX. Default value: 10.
AutoEnvLighting	Flag for automatic lighting. If *AutoEnvLighting* = 1 the lights will be activated/deactivated depending on precipitation, humidity and artificial light (see DPUEnvControl) automatically. Default value: true
AutoEnvThreshold	Threshold for lighting due to precipitation, humidity and artificial light. The threshold calculated using the input values *EnvHumidityOffset*, *EnvHumidityMul*, *EnvArtificialLightMul*, *EnvPrecipitationMul*. The resulting value, which is compared to *AutoEnvThreshold* is also available as output *AutoEnvLightValue*. Default value: 0.5.
AutoEnvDeviation	Deviation for individual random value of *AutoEnvThreshold* for each road user. Default value: 0.05.
AutoBrakeLightControl	Mode for the automatic brake light control of SILAB. Mode = 0 (default): Depending on the driving model that controls a vehicle speed changes do not always result in corresponding acceleration values. Therefore a so called virtual acceleration is calculated within small time segments. The current and the virtual deceleration of a vehicle are both compared to the input variable *AutoBrakeLightThreshold*. If one of both is lower than *AutoBrakeLightThreshold* the brakelights of the vehicle is activated. Mode = 1: The current deceleration of a vehicle is compared to the input variable *AutoBrakeLightThreshold*. A lower deceleration results in activated brakelights.

	Mode = 2: The virtual deceleration (see Mode = 0) of a vehicle is compared to the input variable *AutoBrakeLightThreshold*. A lower deceleration results in activated brakelights. Mode = 3: A weighted average of the latest decaleration is calculated and compared to the input variable *AutoBrakeLightThreshold*. A lower deceleration results in activated brakelights. For this mode the value -3 is a recommend value for *AutoBrakeLightThreshold*. The recommended method is mode 3. The default mode is 0 due to compatibility reasons.
AutoBrakeLightThreshold	Value that has to be reached till the brakelights of a vehicle are activated. For details see input variable *AutoBrakeLightControl*.
EGOBBFront	Distance between center of gravity and front bumper of the simulator mockup. All calculated distances to other simulated vehicles are based on the correct values of the EGOBBXXX variables. [m]
EGOBBLeft	Distance between center of gravity and left border of the simulator mockup. [m]
EGOBBRear	Distance between center of gravity and rear bumper of the simulator mockup. [m]
EGOBBRight	Distance between center of gravity and right border of the simulator mockup. [m]
EGOBrakePedal	Position of the brake pedal in the simulator mockup.
EGOHandBrake	Position of the hand brake in the simulator mockup.
EGOIndicatorLeft	Is the left indicator of the simulator mockup activated? (0 = inactive; 1 = active).
EGOIndicatorRight	Is the right indicator of the simulator mockup activated? (0 = inactive; 1 = active).
EGOLowerBeam	Is the lower beam of simulator mockup activated? (0 = inactive; 1 = active).
EGOODBType	Type of vehicle of the simulator mockup (1 = passenger car)
EGOpitch	Pitch angle of vehicle dynamics of EGO vehicle. [rad]
EGOroll	Roll angle of vehicle dynamics of EGO vehicle. [rad]
EGOv	Velocity of vehicle dynamics of EGO vehicle. [m/s]
EGOX	X position of the EGO vehicle in the global coordinate system. [m]
EGOY	Y positionn of the EGO vehicle in the global coordinate system. [m]
EGOZ	Z position of the EGO vehicle in the global coordinate system. [m]
EGOyaw	Yaw angle of of the EGO vehicle. [rad]
EnableSpray	Flag for spray simulation on wet streets. Default value: false.
EnvArtificialLightMul	For automatic light control (*AutoEnvLighting* = 1): Factor for taking the darkness into account. Darkness is the result of missing ambient and sunlight. Default value: 4.0.
EnvHumidityMul	For automatic light control (*AutoEnvLighting* = 1): Factor for taking into account the humidity. Default value: 1.25.

EnvHumidityOffset	For automatic light control (*AutoEnvLighting* = 1): Offset for taking into account the humidity. The humidity range between [0, *EnvHumidityOffset*] is ignored. Default value: 0.3.
EnvPrecipitationMul	For automatic light control (*AutoEnvLighting* = 1): Factor for taking into account the precipitation. Default value: 1.5.
ExternTrigger[01…25]	Variables to trigger flowpoints externally. For more details see the TRFX reference document.
Fog	The fog density used for automatic lighting, if *AutoFogLighting* is set.
ThreePointHeight	The variable *ThreePointHeight* specifies the method for calculating the orientation of simulated vehicles due to different heights of tire contact points. *ThreePointHeight* = 0: The orientation of a vehicle is defined by the height of the center of gravity ot the vehicle. The height of the center of gravity is defined by the height of the nearest point to the road. *ThreePointHeight = 1 (the default value)*: The orientation of a vehicle is defined by two points of the front axle and one point of the rear axle. The height of the road shoulder and the landscape is also taken into account. This results in much more realistic driving over sidewalks and on the shoulder of a road. The parameter has to be specified for the *DPUTRF* as well as for the *DPUTRFClient*. In both DPUs the same value has to be set.
TrafficDensity	Actual traffic density. A road user will only be activated if its value of ActivateOnDensity (default value is 0.0 – see *Reference Traffic Simulation TRFX* ch. 5.3.3.6) is lower than this input. The default value is 1.0.
UserData[00…39]_In	General purpose inputs. Used by some DPUs of TRF.

2. Parameters:

ODB	Boolean value: Insert road users into the object database ODB? Default: *false (=0)*

3. Outputs:

ActiveController	Number of currently active traffic controller.
ActiveRoadUser	Number of currently active road user.
AutoEnvLightValue	Calculated value for brightness (see input values EnvHumidityOffset, EnvHumidityMul, EnvArtificialLightMul, EnvPrecipitationMul).
DefinedRoadUsers	Total number of defined road users.
NGlobalRoadUser	Number of global road user.

Trafficdata	Current simulation time. This time is transmitted to the DPUTRFClient instances.
UserData[00…39]_Out	General purpose outputs. Used by some DPUs of TRF.

4. Dependencies to other DPUs:

SILAB's traffic simulation consists of the DPUs DPUTRFX, DPUTRFClient and DPUTRFSGE.

These DPUs depend on other DPUs as follows:

DPU	Depends on
DPUTRFX	DPUSCNX
DPUTRFClient	DPUSCNX + Connected to DPUTRFX
DPUTRFSGE	DPUTRFX / DPUTRFClient + DPUSGEWorld

For each dependency, please make sure that a higher *Index* is assigned to the dependent DPU than to its dependencies.

Usually, DPUTRFX is started on the main simulation computer. The graphics computers run DPUTRFClient and DPUTRFSGE each.

For single-computer setups, only DPUTRFX and DPUTRFSGE are required.